SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

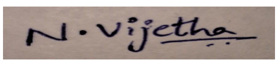
QUESTIONS: 5 **Total Marks:50**

Annexure A
SENARIO BASED QUESTIONS

Criteria	Understanding of Scenario ***(2)***	Identification of Key Issues ***(2)***	Application of Concepts ***(2.5)***	Decision Making ***(2)***	Clarity of Response ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _N.Vijetha (192311258)____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A
Scenario based Questions

Q1. Data Preprocessing in Retail Analytics *(L3, L4)*

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

i. Identify missing values in **Age, Monthly_Spend, and Purchase_Frequency**
ii. Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
iii. Compute the imputed values and justify the selected method

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B. Outlier Detection and Treatment

i. Detect outliers in **Age and Monthly_Spend** using the **Z-score method**
ii. Identify records containing extreme values (e.g., Age = 105, Spend = 500)
iii. Apply suitable treatment methods (**capping / transformation / removal**) and justify

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C. Categorical Data Standardization

i. Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
ii. Convert all entries into a standardized format (**Male / Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics *(L4, L5)*

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
ii. Interpret whether the relationship is **positive or negative**
iii. Explain how covariance helps in identifying feature relationships

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B. Feature Selection

i. Identify redundant attributes using **correlation or domain knowledge**
ii. Select an optimal subset of features for modeling
iii. Justify your selection

C. Data Transformation

i. Standardize the dataset (excluding **Cust_ID and Customer_Class**)
ii. Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics *(L3, L4, L5)*

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

i. Apply **3 bins** for Study_Hours and Test_Score
ii. Calculate bin width and define intervals

B. Equal-Frequency Binning

i. Divide data into **3 bins with equal records**
ii. Assign values accordingly

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High
- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:
200, 300,400, 600, 1000
(a) min-max normalization by setting min = 0 and max = 1
(b) z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis *(L3, L4, L5)*

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

📊 Sample Dataset: Visit Duration

Customer_ID	Visit_Duration (seconds)
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

✅ Tasks

A. Equal-Frequency Binning

i. Divide the dataset into **3 bins** with approximately equal number of records
ii. Assign each value to its respective bin
iii. Analyze how effectively this method groups short and long visit durations

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B. Z-Score Normalization

i. Compute the **mean and standard deviation** of visit duration
ii. Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**

C. Min-Max Normalization

i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

ii. Transform all values accordingly
iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

QUESTION 1

Data Preprocessing in Retail Analytics

Overview

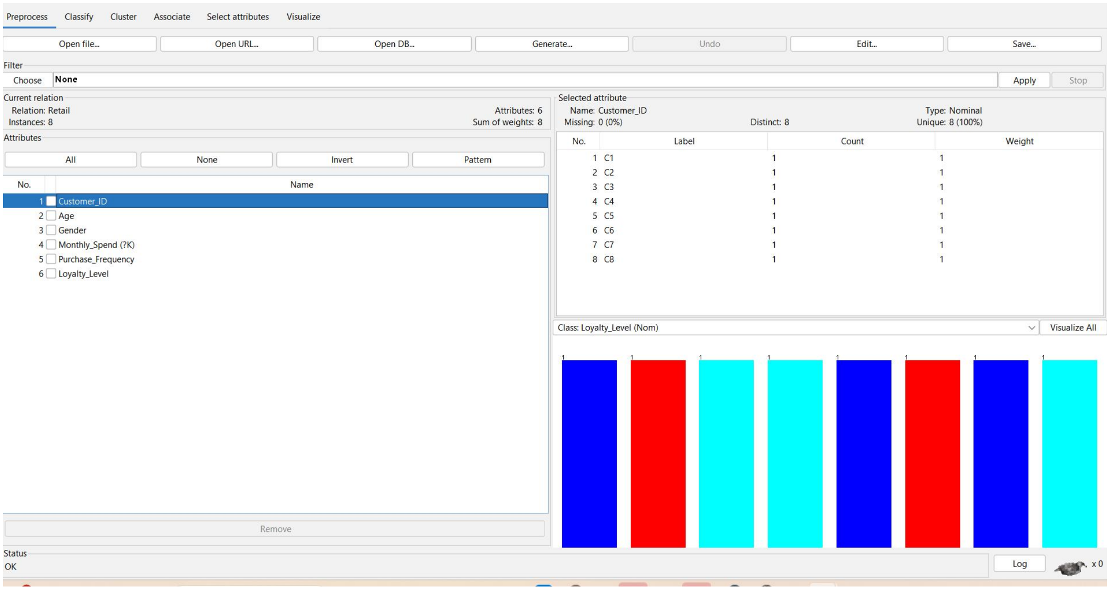
The retail customer purchase dataset includes details such as Customer ID, Age, Gender, Monthly Spend, Purchase Frequency, and Loyalty Level. As the data has been collected from different retail stores and entered manually, it contains missing values, inconsistent text formats, and abnormal values. Data preprocessing is carried out to clean the dataset and improve its quality before performing data mining and analysis.

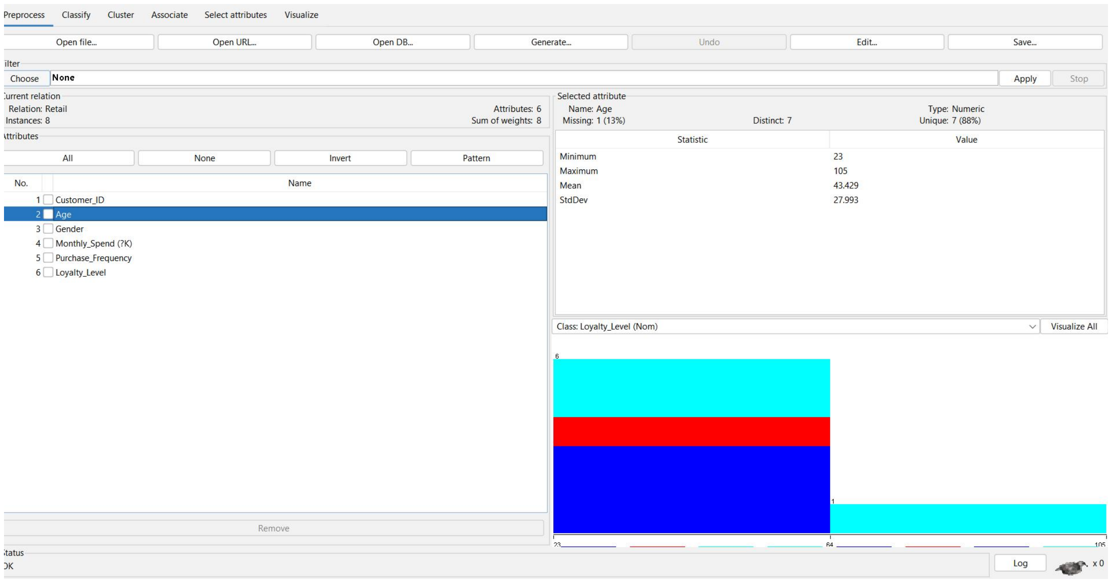
A. Handling Missing Values

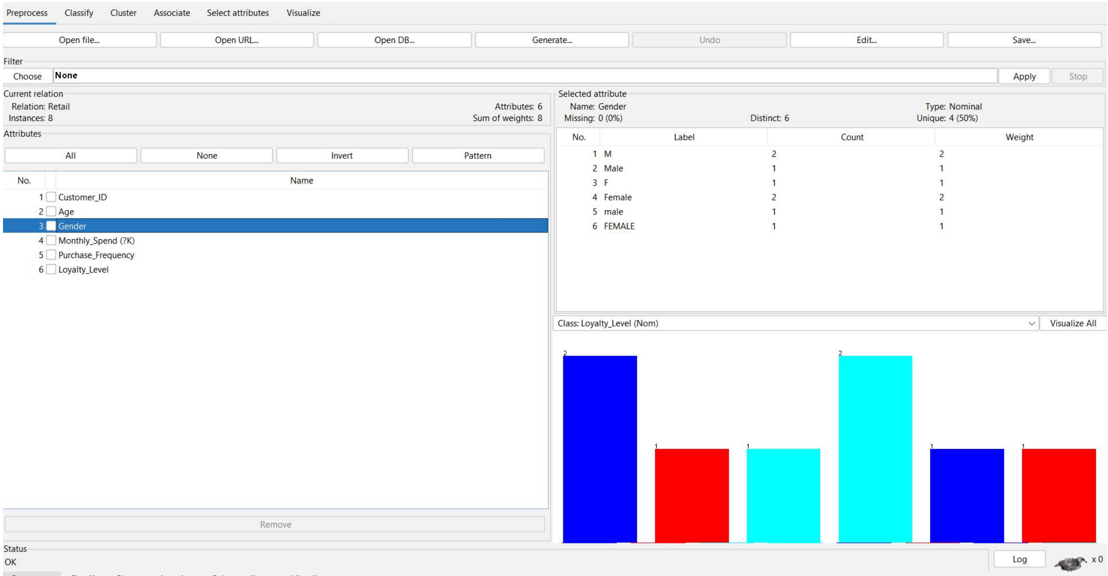
A(i). Detection of Missing Values

The dataset was imported into WEKA using the Preprocess tab. Each attribute was checked to identify missing entries. The missing values were found in the following attributes.

Attribute	Customer ID	Missing Entry
Age	C2	?
Monthly_Spend	C3	?
Purchase_Frequency	C5	?



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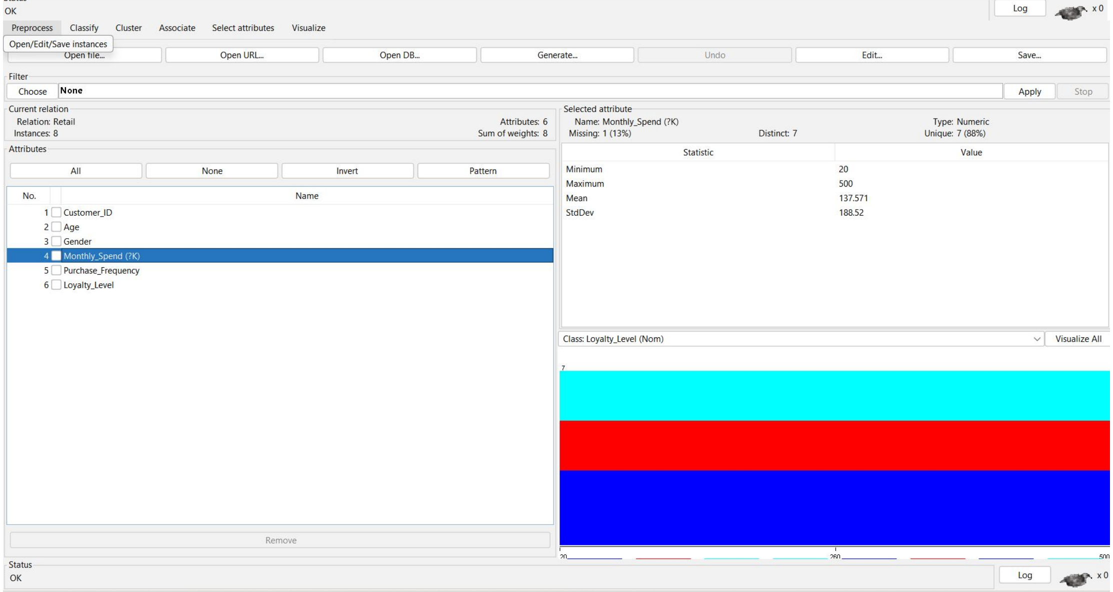
Current relation
Relation: Retail
Instances: 8
Attributes: 6
Sum of weights: 8
Attributes
All
None
Invert
Pattern
No.
Name
1 Customer_ID
2 Age
3 Gender
4 Monthly_Spend (?K)
5 Purchase_Frequency
6 Loyalty_Level
Remove
Selected attribute
Name: Gender
Missing: 0 (0%)
Distinct: 6
Type: Nominal
Unique: 4 (50%)
No. Label Count Weight
1 M 2 2
2 Male 1 1
3 F 1 1
4 Female 2 2
5 male 1 1
6 FEMALE 1 1
Class: Loyalty_Level (Nom)
Visualize All
Status
OK
Log
x 0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Attributes
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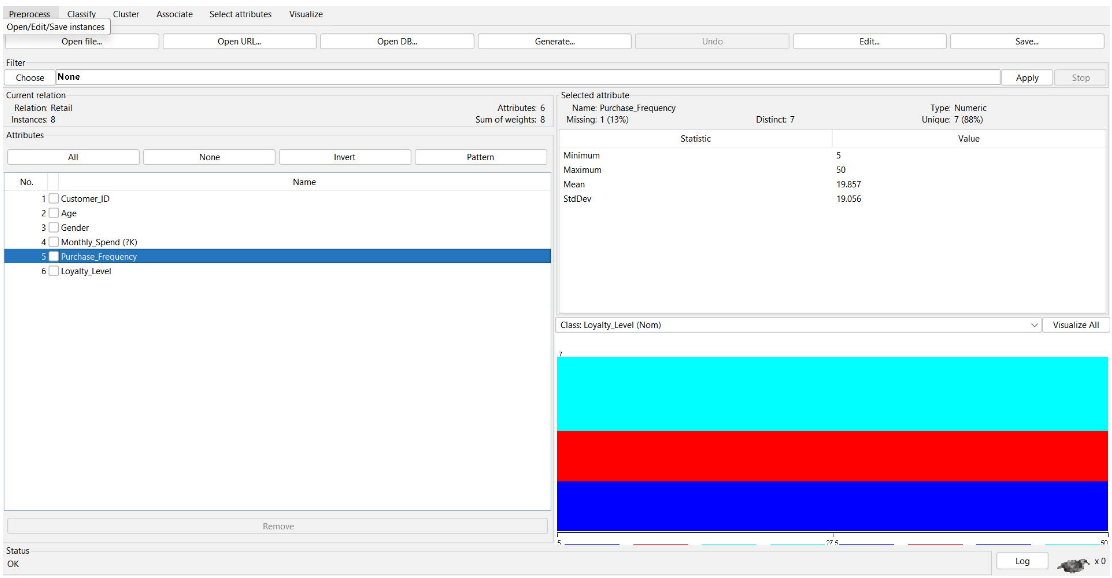
Pattern
No.
Name
1 Customer_ID
2 Age
3 Gender
4 Monthly_Spend (?K)
5 Purchase_Frequency
6 Loyalty_Level
Remove
Selected attribute
Name: Monthly_Spend (?K)
Missing: 1 (13%)
Distinct: 7
Type: Numeric
Unique: 7 (88%)
Statistic Value
Minimum 20
Maximum 500
Mean 137.571
StdDev 188.52
Class: Loyalty_Level (Nom)
Visualize All
Status
OK
Log
x 0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Attributes
All
None
Invert
Pattern
No.
Name
1 Customer_ID
2 Age
3 Gender
4 Monthly_Spend (?K)
5 Purchase_Frequency
6 Loyalty_Level
Remove
Selected attribute
Name: Purchase_Frequency
Missing: 1 (13%)
Distinct: 7
Type: Numeric
Unique: 7 (88%)
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Log
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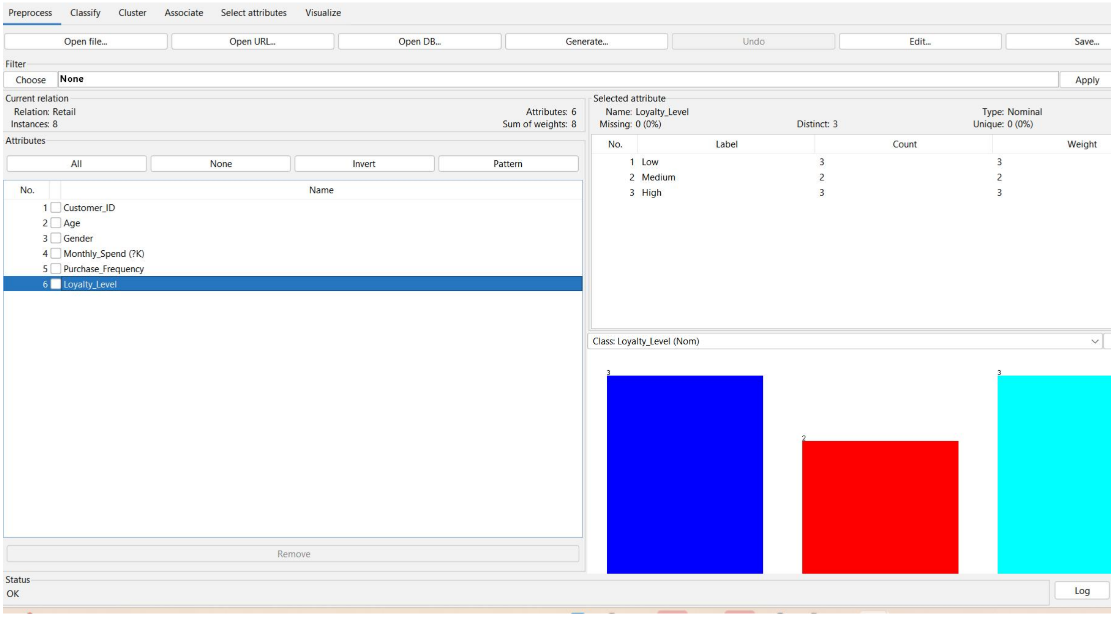


A(ii). Imputation Techniques Used

Missing values can be handled using different imputation methods. Two commonly used techniques are described below.

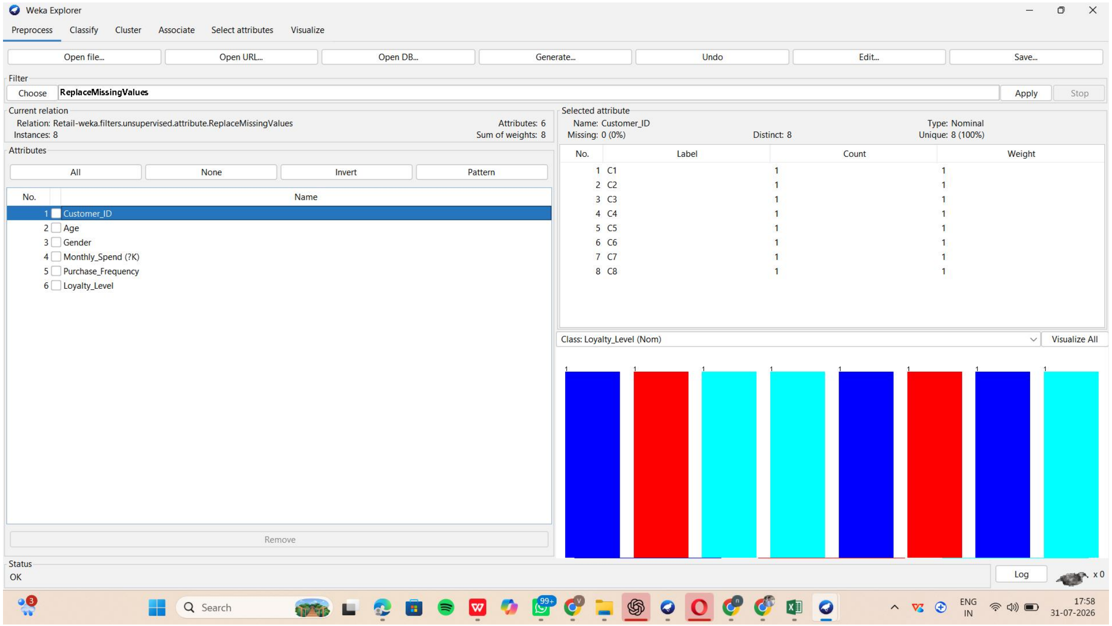
Median Imputation

In this technique, the missing numerical value is replaced with the median value of that attribute. Since the median is not significantly affected by extreme values, it is suitable for datasets that contain outliers.

k-Nearest Neighbor (k-NN) Imputation

This technique estimates the missing value by considering records that have similar characteristics. The missing value is calculated based on the values of the nearest neighboring instances, resulting in a more reliable estimation.

For this experiment, the ReplaceMissingValues filter in WEKA was used to handle the missing values present in the dataset.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Apply
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Current relation
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Instances: 8
Attributes: 6
Sum of weights: 8
Attributes
All
None
Invert
Pattern
No.
Name
1 Customer_ID
2 Age
3 Gender
4 Monthly_Spend (?K)
5 Purchase_Frequency
6 Loyalty_Level
Remove
Selected attribute
Name: Customer_ID
Missing: 0 (0%)
Distinct: 8
Type: Nominal
Unique: 8 (100%)
No. Label Count Weight
1 C1 1 1
2 C2 1 1
3 C3 1 1
4 C4 1 1
5 C5 1 1
6 C6 1 1
7 C7 1 1
8 C8 1 1
Class: Loyalty_Level (Nom)
Visualize All
Status
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Log
x 0
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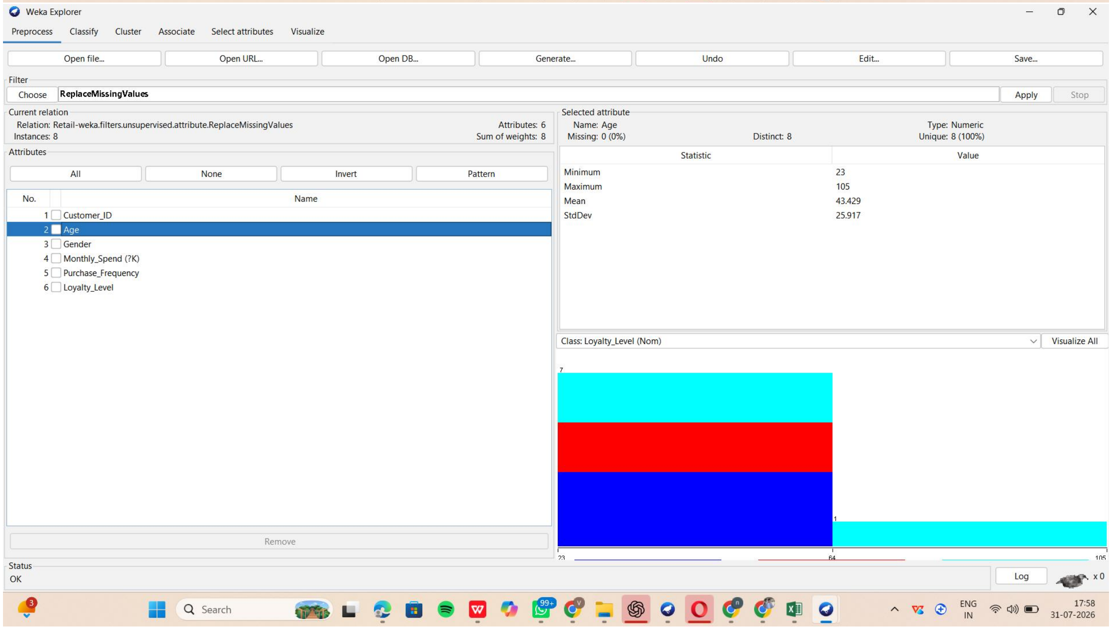
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Attributes
All
None
Invert
Pattern
No.
Name
1 Customer_ID
2 Age
3 Gender
4 Monthly_Spend (?K)
5 Purchase_Frequency
6 Loyalty_Level
Remove
Selected attribute
Name: Age
Missing: 0 (0%)
Distinct: 8
Type: Numeric
Unique: 8 (100%)
Statistic Value
Minimum 23
Maximum 105
Mean 43.429
StdDev 25.917
Class: Loyalty_Level (Nom)
Visualize All
Status
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Log
x 0
Search
ENG
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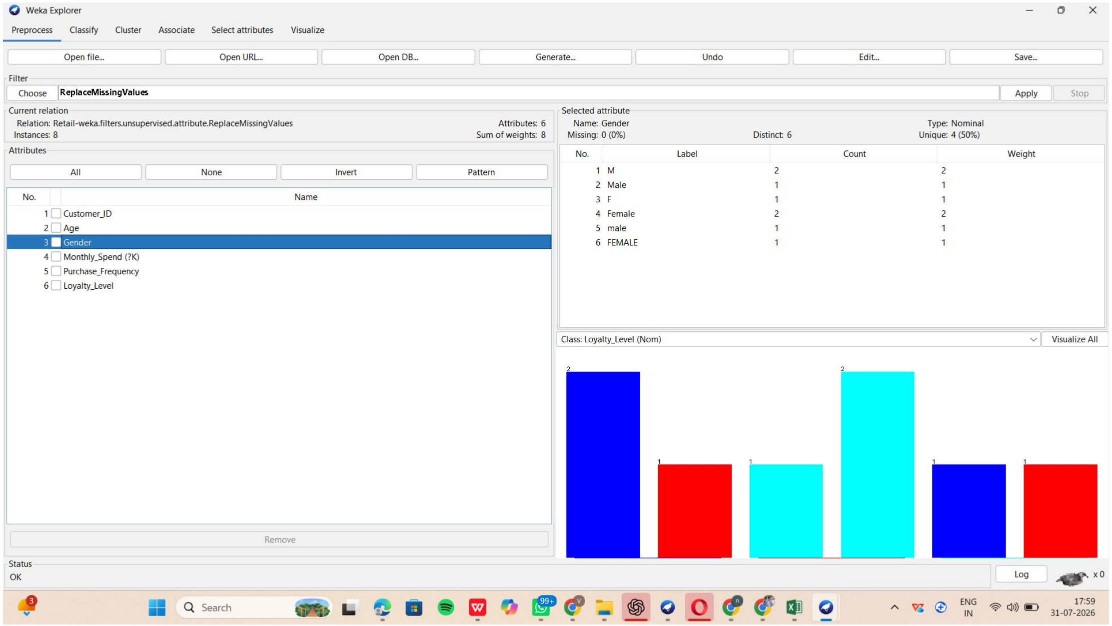
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Instances: 8
Attributes: 6
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Attributes
All
None
Invert
Pattern
No.
Name
1 Customer_ID
2 Age
3 Gender
4 Monthly_Spend (?K)
5 Purchase_Frequency
6 Loyalty_Level
Remove
Selected attribute
Name: Gender
Missing: 0 (0%)
Distinct: 6
Type: Nominal
Unique: 4 (50%)
No. Label Count Weight
1 M 2 2
2 Male 1 1
3 F 1 1
4 Female 2 2
5 male 1 1
6 FEMALE 1 1
Class: Loyalty_Level (Nom)
Visualize All
Status
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Log
x 0
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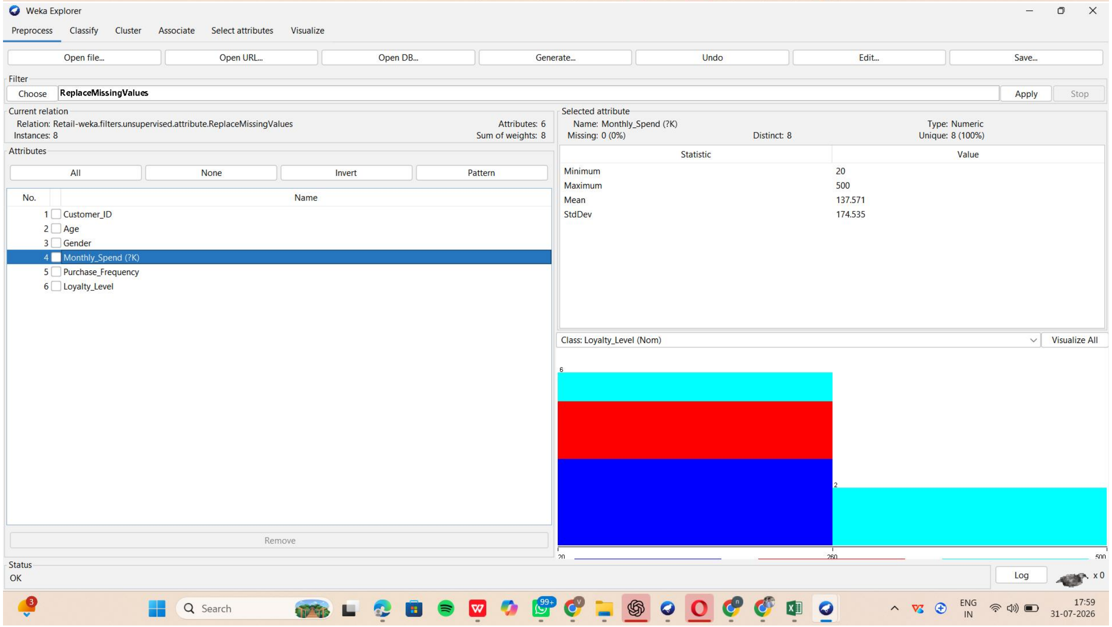
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4 Monthly_Spend (?K)
5 Purchase_Frequency
6 Loyalty_Level
Remove
Selected attribute
Name: Monthly_Spend (?K)
Missing: 0 (0%)
Distinct: 8
Type: Numeric
Unique: 8 (100%)
Statistic Value
Minimum 20
Maximum 500
Mean 137.571
StdDev 174.535
Class: Loyalty_Level (Nom)
Visualize All
Status
OK
Log
x 0
Search
ENG
IN
17:59
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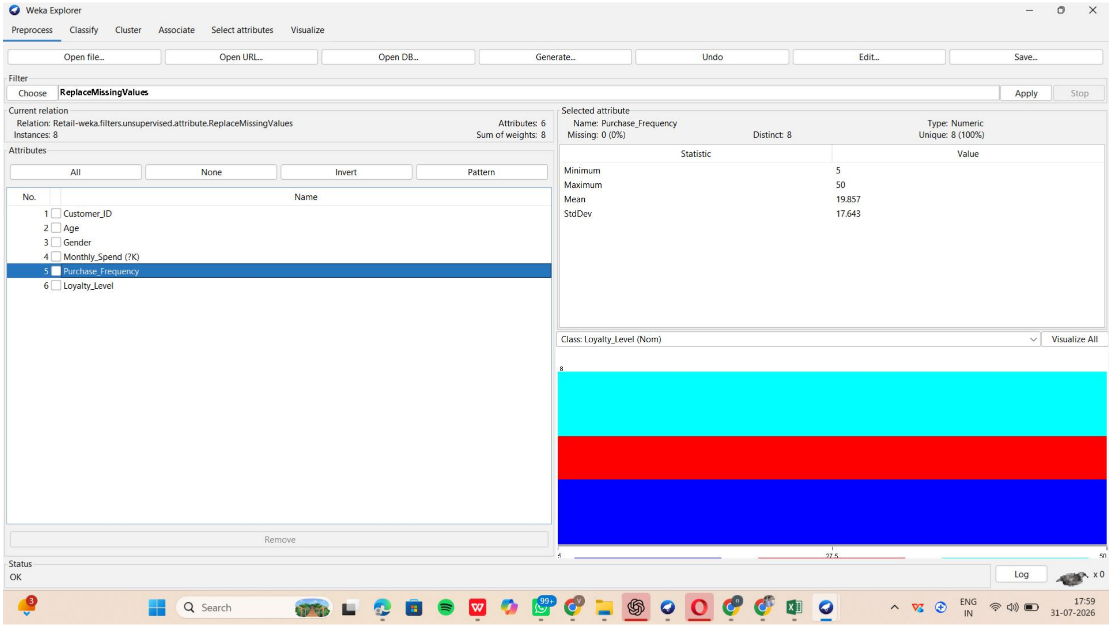
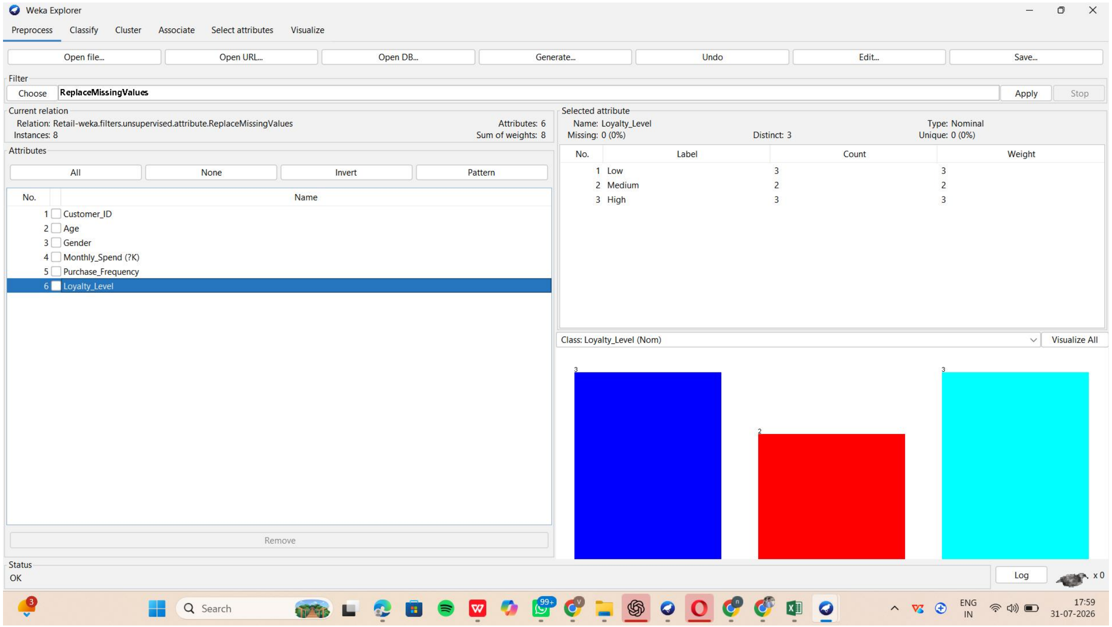




A(iii). Imputed Values and Justification

After applying the preprocessing filter, all missing values were successfully replaced, resulting in a complete dataset. Handling missing values prevents incomplete records from affecting the performance of future analysis.

Median imputation is effective when the dataset contains outliers because it is resistant to extreme values. k-NN imputation is useful when the missing value depends on the characteristics of similar records, thereby maintaining the relationships among attributes.

B. Outlier Detection and Treatment

B(i). Outlier Identification Using the Z-Score Method

The attributes Age and Monthly_Spend were analyzed to identify abnormal values. The Z-score technique measures how far each value is from the average of the dataset. Values with very high positive or negative Z-scores are considered outliers.

B(ii). Records Containing Extreme Values

The following observations were identified as extreme values because they are considerably larger than the remaining records.

Customer ID	Attribute	Extreme Value
C4	Age	105
C4	Monthly_Spend	500
C8	Monthly_Spend	300

These records require appropriate treatment before building predictive models.

B(iii). Outlier Treatment

Several techniques can be used to handle outliers:

Capping: Limits extreme values to a predefined upper or lower threshold.

Transformation: Reduces the impact of large values using mathematical functions such as logarithmic transformation.

Removal: Eliminates records containing incorrect or invalid values from the dataset.

For this dataset, capping is considered the most suitable approach because it minimizes the influence of extreme values while retaining all customer records for analysis.

C. Categorical Data Standardization

C(i). Identification of Inconsistent Gender Values

The Gender attribute contains multiple representations of the same category, leading to inconsistency within the dataset.

Existing Values

M

Male

male

F

Female

FEMALE

Such inconsistencies may produce duplicate categories during data processing and affect analysis accuracy.

C(ii). Standardization of Gender Values

To maintain consistency, all gender entries were converted into a common format.

Existing Value	Standardized Value
M	Male
Male	Male
male	Male
F	Female
Female	Female
FEMALE	Female

After standardization, the dataset contains only two valid categories: Male and Female. This improves data consistency, simplifies analysis, and reduces errors during data mining and machine learning.

QUESTION 2

Covariance and Data Reduction in E-commerce Analytics

Overview

The e-commerce dataset consists of customer-related information, including Age, Income, Purchase Value, Credit Score, Number of Orders, Average Balance, EMI, and Customer Class. Before applying machine learning techniques, the dataset should be analyzed and preprocessed. This involves measuring relationships between variables, selecting important features, and transforming numerical data into a common scale to improve model efficiency and prediction accuracy.

A. Covariance Analysis

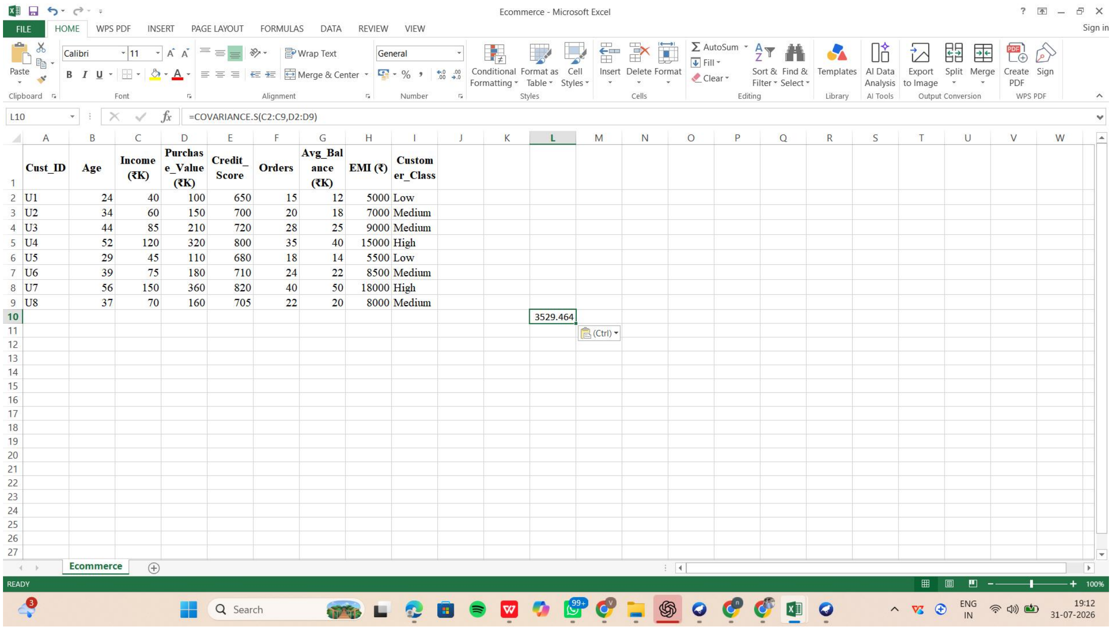
A(i). Computing Covariance

Covariance was calculated to determine the relationship between the Income and Purchase_Value attributes. Microsoft Excel was used to perform this calculation with the COVARIANCE.S() function.

Formula Used:

=COVARIANCE.S(C2:C9,D2:D9)

Calculated Covariance Value: 3529.464



Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

A(ii). Analysis of Covariance

The covariance obtained is 3529.464, which is greater than zero. This positive value indicates that the two variables have a direct relationship. As the income of customers increases, their purchase value also tends to increase, showing a positive association between these attributes.

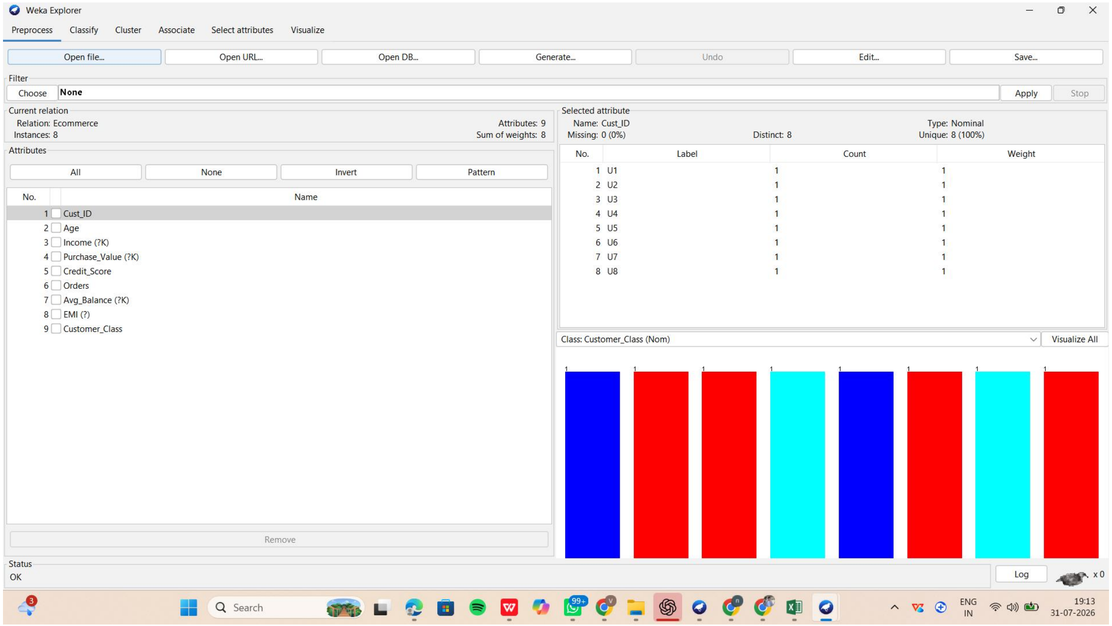
A(iii). Significance of Covariance

Covariance is a useful statistical technique for identifying how two numerical variables are related. It provides insight into whether attributes change together and assists in recognizing correlated variables. This information is valuable when selecting features for data mining and predictive modeling.

B. Feature Selection

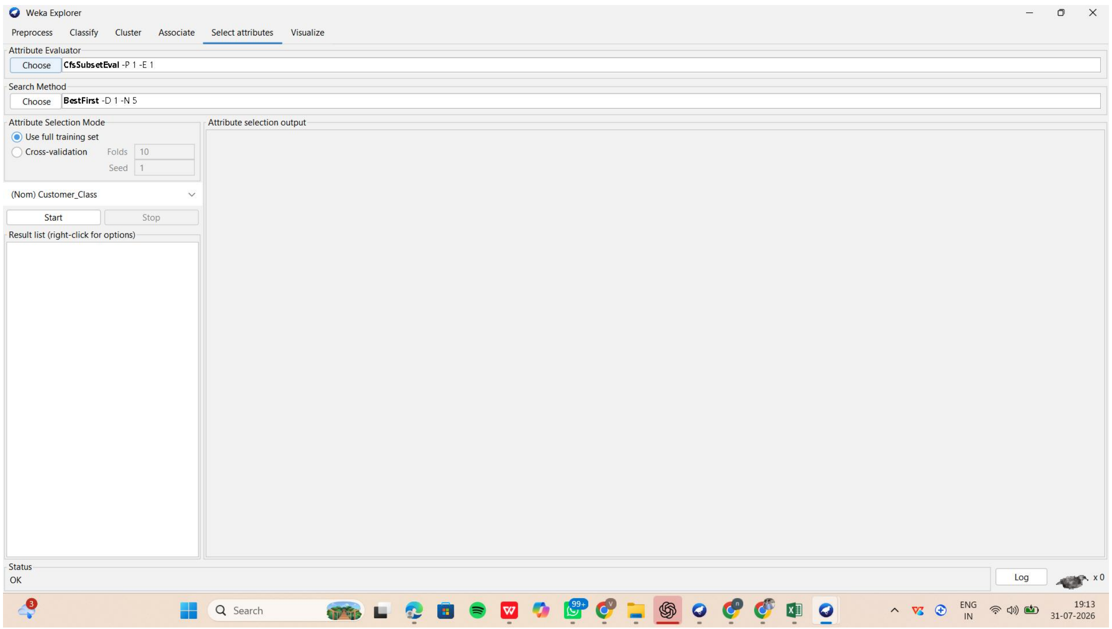
B(i). Selecting Relevant Attributes

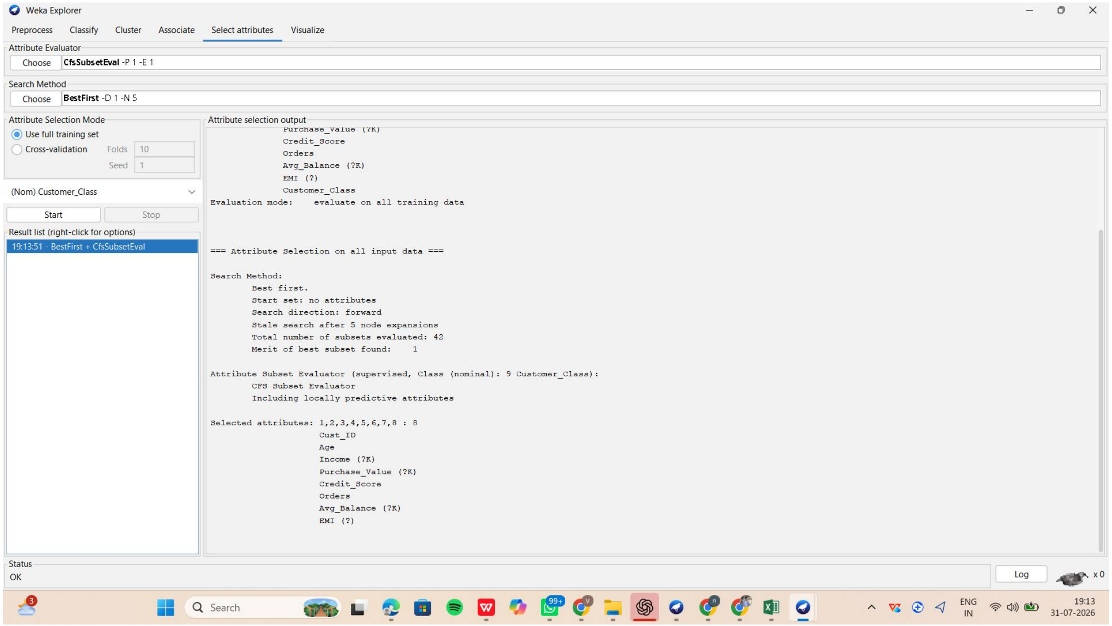
The dataset was opened in the WEKA Explorer, and feature selection was performed using the Select Attributes tab. The CfsSubsetEval evaluator, along with the BestFirst search method, was used to identify the most useful attributes while eliminating redundant ones.



B(ii). Selected Attributes

After executing the feature selection process, WEKA identified the attributes that contribute most effectively to customer classification. These selected features improve the efficiency of the model by removing unnecessary or highly correlated attributes.





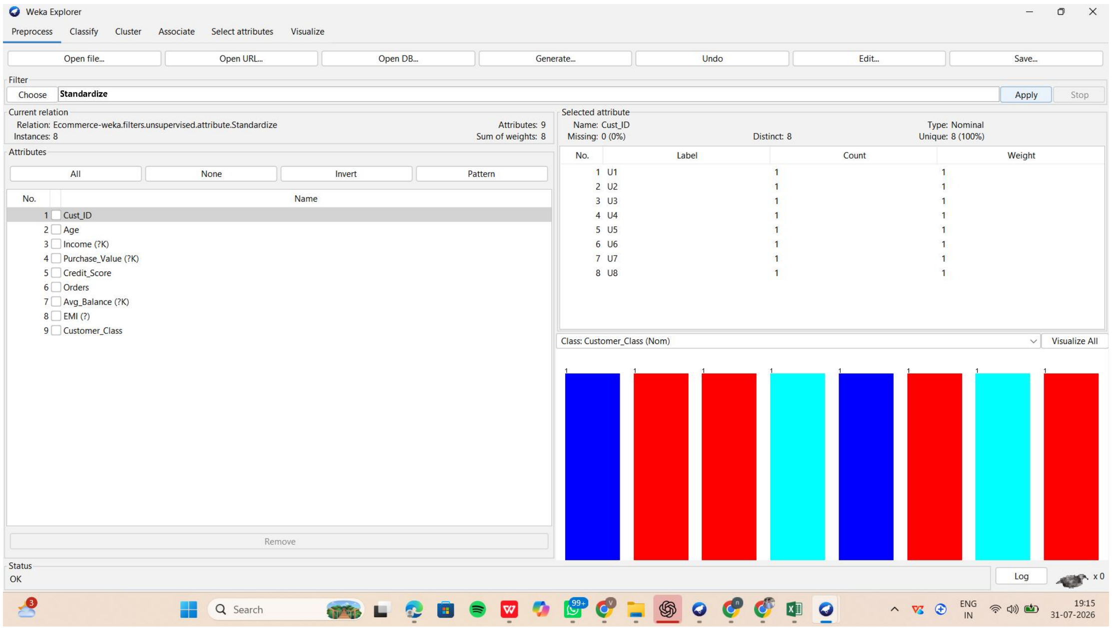
B(iii). Benefits of Feature Selection

Feature selection simplifies the dataset by removing irrelevant and duplicate information. This reduces training time, lowers computational requirements, and improves the overall performance and reliability of machine learning models.

C. Data Transformation

C(i). Applying Standardization

To ensure that all numerical attributes are measured on a common scale, the Standardize filter available in the Preprocess tab of WEKA was applied. Identifier fields such as Cust_ID and the target attribute Customer_Class were not included in the transformation process.



C(ii). Purpose of Standardization

Standardization converts numerical attributes into values with a common distribution by adjusting the mean and standard deviation. This prevents variables with larger numerical ranges from influencing the model more than others and improves the performance of algorithms that rely on distance or numerical scaling.

Conclusion

The e-commerce dataset was successfully preprocessed using covariance analysis, feature selection, and standardization techniques. Covariance revealed a positive relationship between Income and Purchase Value. Feature selection retained the most informative attributes for classification, while standardization converted the numerical attributes to a uniform scale. These preprocessing steps improve data consistency, reduce redundancy, and prepare the dataset for accurate machine learning analysis.

QUESTION 3

Data Transformation and Discretization in Education Analytics

Overview

The given education dataset contains information about students, including Study Hours, Test Score, Attendance Percentage, and Performance. Since Study Hours and Test Score are continuous numerical values, they can be converted into discrete groups using binning techniques. Data transformation improves data readability, simplifies analysis, and helps in making better academic predictions.

A. Equal-Width Binning

A(i). Dividing Study_Hours and Test_Score into Three Equal Bins

Study_Hours

The minimum study hours recorded are 2, while the maximum value is 11.

Range = 11 − 2 = 9

Number of bins = 3

Bin Width = Range ÷ Number of Bins = 9 ÷ 3 = 3

The following intervals were created:

Bin	Study_Hours Interval
Bin 1	2 – 5
Bin 2	5 – 8
Bin 3	8 – 11

Study_Hours Bin Assignment

Student ID	Study_Hours	Assigned Bin
S1	2	Bin 1
S2	3	Bin 1
S3	4	Bin 1
S4	5	Bin 1
S5	6	Bin 2
S6	7	Bin 2
S7	8	Bin 2
S8	9	Bin 3
S9	10	Bin 3
S10	11	Bin 3

Test_Score

The minimum test score is 40 and the maximum score is 100.

Range = 100 − 40 = 60

Number of bins = 3

Bin Width = 60 ÷ 3 = 20

The score intervals are:

Bin	Test Score Interval
Bin 1	40-6-
Bin 2	61-80
Bin 3	81-100

Test_Score Bin Assignment

Student ID	Test Score	Assigned Bin
S1	40	Bin 1
S2	50	Bin 1
S3	55	Bin 1
S4	65	Bin 2
S5	75	Bin 2
S6	85	Bin 3
S7	90	Bin 3
S8	95	Bin 3

S9	98	Bin 3
S10	100	Bin 3

B. Equal-Frequency Binning

In equal-frequency binning, the dataset is divided so that each group contains nearly the same number of observations. Since the dataset contains 10 student records, they are distributed into three groups.

Study_Hours

Bin	**Values Included**
Bin 1	2,3,4
Bin 2	5,6,7
Bin 3	8,9,10,11

Test_Score

Bin	**Values Included**
Bin 1	40,50,55
Bin 2	65,75,85
Bin 3	90,95,98,100

This method ensures that every bin contains almost an equal number of records, making the distribution more balanced.

C. Concept Hierarchy Construction

Concept hierarchies convert detailed numerical values into broader descriptive categories.

Study_Hours Categories

Study Hours	Level
2-4	Low
5-7	Medium
8-11	High

Test_Score Categories

Test Score	**Level**
40-60	Poor
61-80	Average
81-100	Excellent

Using these categories makes it easier to compare student performance and interpret the dataset.

QUESTION 4

Normalization Techniques

Introduction

Normalization is an important data preprocessing method used to convert numerical values into a common scale. It reduces the effect of differences in value ranges and ensures that no single attribute has an unfair influence during analysis. Applying normalization improves the accuracy and efficiency of data mining and machine learning algorithms.

A. Min-Max Normalization

Given Data

200, 300, 400, 600, 1000

Formula

$$\text{Normalized Value} = \frac{X - \text{Minimum}}{\text{Maximum} - \text{Minimum}}$$

Calculation

Minimum Value = 200

Maximum Value = 1000

Difference (Range) = 1000 − 200 = 800

Original Value	Calculation	Normalized Value
200	(200 − 200) / 800	0.000
300	(300 − 200) / 800	0.125
400	(400 − 200) / 800	0.250
600	(600 − 200) / 800	0.500
1000	(1000 − 200) / 800	1.000

Observation

Using Min-Max Normalization, all values are converted to the range 0 to 1. The smallest value becomes 0, while the largest value becomes 1, and the remaining values are proportionally scaled between them.

B. Z-Score Normalization

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Where:

μ (Mean) = 500

σ (Standard Deviation) ≈ 316.23

Normalized Values

Original Value	Z-Score
200	-0.95
300	-0.63
400	-0.32
600	0.32
1000	1.58

Observation

Z-Score Normalization converts each value based on its distance from the mean. Negative values indicate observations below the mean, whereas positive values represent observations above the mean. This method is useful when data needs to be standardized without restricting it to a fixed range.

Summary

In this exercise, the given dataset was normalized using Min-Max Normalization and Z-Score Normalization. Min-Max Normalization rescaled the data between 0 and 1, while Z-Score Normalization expressed each value in terms of its deviation from the mean. Both techniques help create consistent datasets, reduce the impact of varying numerical scales, and improve the effectiveness of data analysis and predictive modeling.

QUESTION 5

Data Transformation in Online Shopping Behavior Analysis

Introduction

Online shopping platforms record the amount of time customers spend browsing products. Since some users spend only a few seconds while others remain on the website for a long time, the data becomes unevenly distributed. Before applying clustering techniques, the visit duration data should be transformed using suitable preprocessing methods. In this experiment, Equal-Frequency Binning, Z-Score Normalization, and Min-Max Normalization are applied to prepare the dataset for analysis.

A. Equal-Frequency Binning

A(i). Creating Three Equal-Frequency Groups

The dataset contains 10 customer records. These records are divided into 3 groups, ensuring that each group contains nearly the same number of observations.

Visit Duration (Seconds)

15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600

Binning Result

Bin	Visit Duration (Seconds)
Bin 1	15, 30, 45
Bin 2	60, 120, 300
Bin 3	600, 900, 1800, 3600

A(ii). Customer Bin Allocation

Customer ID	Visit Duration (Seconds)	Assigned Bin
C1	15	Bin 1
C2	30	Bin 1
C3	45	Bin 1
C4	60	Bin 2
C5	120	Bin 2
C6	300	Bin 2
C7	600	Bin 3
C8	900	Bin 3
C9	1800	Bin 3
C10	3600	Bin 3

A(iii). Interpretation

Equal-frequency binning distributes records evenly among the bins. Customers with shorter browsing times are placed in the first group, moderate browsing times in the second group, and customers with long browsing durations in the final group. This grouping helps simplify customer behavior analysis before clustering.

B. Z-Score Normalization

B(i). Mean and Standard Deviation

Input Data

15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600

The average visit duration is calculated as:

$$\mu = \frac{15 + 30 + 45 + 60 + 120 + 300 + 600 + 900 + 1800 + 3600}{10} = 747$$

Mean (μ) = 747 seconds

The standard deviation of the dataset is:

Standard Deviation (σ) ≈ 1114.08 seconds

B(ii). Standardized Values

The following equation is used:

$$Z = \frac{X - \mu}{\sigma}$$

Visit Duration	Z-Score
15	-0.66
30	-0.64
45	-0.63
60	-0.62
120	-0.56
300	-0.40

600	-0.13
900	0.14
1800	0.94
3600	2.56

B(iii). Interpretation

Z-score normalization measures how far each value lies from the average. Smaller visit durations produce negative Z-scores, whereas larger durations produce positive Z-scores. Extremely large values such as 1800 and 3600 seconds have the highest positive Z-scores, making them easy to identify during analysis.

C. Min-Max Normalization

C(i). Scaling the Data

The Min-Max normalization formula is:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Where:

Minimum Visit Duration = 15 seconds

Maximum Visit Duration = 3600 seconds

Range = 3585 seconds

C(ii). Normalized Dataset

Visit Duration	Normalized Value
15	0.000
30	0.004
45	0.008
60	0.013
120	0.029
300	0.080
600	0.163

900	0.247
1800	0.498
3600	1.000

C(iii). Comparison of the Two Methods

Min-Max normalization rescales every value into the interval 0 to 1, making the data suitable for algorithms that require normalized input. Z-score normalization, however, standardizes the data based on the mean and standard deviation, making it easier to detect unusually large or small values. The choice between the two methods depends on the type of analysis and the characteristics of the dataset.

Conclusion

The visit duration data was successfully transformed using three preprocessing techniques. Equal-frequency binning grouped customers according to similar browsing durations, Z-score normalization standardized the data by considering the mean and standard deviation, and Min-Max normalization converted the values into a common range of 0 to 1. These transformations improve the quality of the dataset and make it more suitable for clustering and customer behavior analysis.